

EDUCATION

Department of Computer Science and Engineering, Yonsei University Seoul, Korea
MSc/PhD Student in Computer Science 2022 - Present

- Advisor: Prof. Youngsok Kim
- Research area: System Software, Database System, Processing-in-Memory

Department of Computer Science and Engineering, Yonsei University Seoul, Korea
BSc in Computer Science 2017 - 2022

PUBLICATIONS

1. Sungchul Lee, **Suhyun Lee**, Hanna Cha, Chaemin Lim, and Youngsok Kim. GJP: Fast and Efficient Join-Project Query Processing on GPUs Using Density-Aware Matrix Multiplication. *In Proc. ACM on Management of Data (SIGMOD)*, 2027.
2. Jinwoo Choi, **Suhyun Lee**, and Youngsok Kim. RecoNN: A Resource Contention-Aware Runtime for Latency-Sensitive On-Device Inference. *In Proc. 44th IEEE International Conference on Computer Design (ICCD)*, 2026.
3. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Kwanghyun Park, Jinho Lee, Joonsung Kim, and Youngsok Kim. FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables. *In Proc. 42nd IEEE International Conference on Data Engineering (ICDE)*, 2026.
4. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Joonsung Kim, Jinho Lee, Youngsok Kim. DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP. *In Proc. 34th International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2025.
5. **Suhyun Lee**, Chaemin Lim, Jinwoo Choi, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim. SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs. *In Proc. ACM on Management of Data (SIGMOD)*, 2025.
6. Chaemin Lim, **Suhyun Lee**, Jinwoo Choi, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim. Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs. *In Proc. ACM on Management of Data (SIGMOD)*, 2023.
7. Jinwoo Choi, Jaeyeon Kim, Chaemin Lim, **Suhyun Lee**, Jinho Lee, Dokyung Song, and Youngsok Kim. GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware. *In Proc. 23rd ACM/IFIP International Middleware Conference (Middleware)*, 2022.
8. Jounghoo Lee, Yeonan Ha, **Suhyun Lee**, Jinyoung Woo, Jinho Lee, Hanhwi Jang, and Youngsok Kim. GCoM: a detailed GPU core model for accurate analytical modeling of modern GPUs. *In Proc. 49th Annual International Symposium on Computer Architecture (ISCA)*, 2022.

PROJECTS

Optimizing Database Systems Using Processing-in-Memory Hardware
National Research Foundation of Korea (NRF) 2024 - 2026

Exploring Quality-Aware NN Scheduler
LG Electronics 2024 - 2025

AWARDS AND HONORS

- **Scholarship** NRF Fellowship Program for Doctoral Students. 2024.06
- **Best Poster Award** Computer System Society Academic Conference 2023.02

EXPERIENCE	Teaching Assistant: <i>Computer Architecture (CSI3102),</i>	2023S
	<i>Logic Circuit Design (CSI2111),</i>	2022F
	<i>Information Security (CSI4109)</i>	2022S
SKILLS	Languages: Korean, English.	
	Programming: C/C++, Python.	